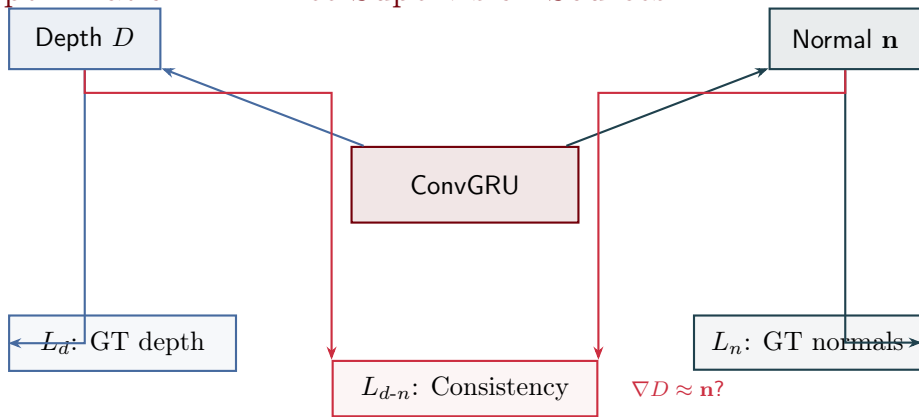


Joint Optimization — Three Supervision Sources



Converts depth gradients to pseudo-normals; penalizes disagreement

Key insight: Only $\sim 20\text{K}$ outdoor images have GT normals. The consistency loss L_{d-n} lets the model learn normals from depth-only datasets by enforcing geometric agreement.